

Group 1 Report: Selective Contrastive Learning for Weakly Supervised Affordance Grounding

Paper title: *Selective Contrastive Learning for Weakly Supervised Affordance Grounding* (ICCV 2025)

1. Git Clone Status

The repository is cloned at /root/workspace/andycho/CV/SelectiveCL. It is on branch main with latest local commit hash 8a50d38. It has both the original SelectiveCL remote (<https://github.com/hynnnsk/SelectiveCL.git>) and our project remote ([git@github.com:airman-and/DSC3033_WSAG.git](https://github.com/airman-and/DSC3033_WSAG.git)). The working directory also contains our analysis notes, extracted figures, notebooks, checkpoints, logs, and generated run outputs.

2. Code Execution Status

We were able to run the code on an NVIDIA Quadro RTX 8000 GPU server using the pinned selectivecl conda environment: Python 3.7.16, PyTorch 1.9.0, CUDA Toolkit 11.1.1, torchvision 0.10.0, OpenAI CLIP, fast-pytorch-kmeans, pycocotools, OpenCV, and Matplotlib. The script run_selectivecl_all_gpu3.sh targets server GPU index 3, an NVIDIA Quadro RTX 8000, and exposes it as logical GPU 0. AGD20K is located at /root/workspace/andycho/CV/AGD20K.

The full AGD20K run completed on May 8, 2026. Both 15-epoch training jobs finished for Seen and Unseen, and official checkpoints were tested:

Split / output	KLD	SIM	NSS
Seen official	1.142	0.415	1.303
Seen refined ego-ego	1.124	0.433	1.280
Unseen official	1.287	0.378	1.377
Unseen refined ego-ego	1.243	0.405	1.368

A short smoke-training demo also ran for two training steps and two evaluation steps. Main issues were environment compatibility and dataset availability: the code requires an older Python/CUDA/PyTorch stack, long API-shell background jobs needed a persistent terminal session, and HICO-IIF has not been evaluated because /DATA/HICO-IIF is absent.

3. Task Division Plan

- Christian Zantua: repository/environment setup, dependency checks, checkpoint management, reproducible scripts, and GPU/run-log tracking.
- Hwang Hee Seok: paper and source-code understanding, especially mapping the method to models/locate.py, loss/loss.py, and the data pipeline.
- Cho Hyun Young: evaluation, KLD/SIM/NSS collection, qualitative visualizations from test.py, and final report/demo notebook preparation.

4. Paper Understanding

SelectiveCL studies weakly supervised affordance grounding: localizing object parts that support an action, such as grasping, cutting, or pouring, without dense masks during training. It learns from image-level affordance labels and paired egocentric/exocentric images.

The method first uses CLIP/ClearCLIP-style patch-text similarity to find action-associated object regions. It then uses DINO ViT-S/16 patch descriptors and self-attention to reason about parts. If an exocentric part cue is reliable, selective prototypical contrastive learning uses it as part-level supervision; otherwise, the model falls back to object-level cues to avoid noisy pseudo-parts. Selective pixel contrastive learning further separates egocentric patch features into positive and negative groups using exocentric object affinity.

In code, the main training logic is in models/locate.py::Net.forward; inference is simpler and uses only the egocentric image through Net.test_forward. Dense GT heatmaps are not used in the training objective. GT masks are used only for epoch-end validation and final testing with KLD, SIM, and NSS.

5. Training and Evaluation Plans

Hardware: local NVIDIA Quadro RTX 8000 GPU server with the pinned selectivecl environment. Colab/Colab Pro is a fallback; the notebooks support AGD20K mounted from Google Drive.

Models and datasets: evaluate official SelectiveCL checkpoints and our reproduced checkpoints on AGD20K-Seen and AGD20K-Unseen. HICO-IIF is future work if the dataset and test path are prepared.

Metrics: reproduce KLD, SIM, and NSS for raw ego predictions and CLIP-refined outputs (refined_CLIP_ego_ego, refined_CLIP_ego_mean). KLD is lower-is-better; SIM and NSS are higher-is-better. Training loss/classification accuracy logs and qualitative heatmap overlays will be used as sanity checks.